1 Linear Functionals

Definition 1.1 (Linear Functional)**.** Let V be a vector space over F. An element of $\mathscr{L}(V,F)$ is called a *linear functional* on V.

Exercise 1.2. Fix $a_1,\ldots,a_n \in F$ and define $f : F^n \to F$ by $f(x_1,\ldots,x_n) = a_1x_1 + \cdots + a_nx_n$.

(a) Show that f is a linear functional on F^n and find the matrix of f relative to the standard bases of F^n and F.

(b) Show that every linear functional on F^n is of this form, for some scalars $a_1,\ldots,a_n$.

Exercise 1.3. Let $V = \mathscr{P}(F)$ and fix $t \in F$. Show that the *evaluation map* $\text{ev}_t : V \to F$ defined by $\text{ev}_t(p) = p(t)$ is a linear functional.

Exercise 1.4. Fix a closed interval $[a,b]$ of the real line and let $V = \mathscr{C}([a,b])$, the vector space of continuous functions on $[a,b]$. Show that $f : V \to \mathbb{R}$ defined by $f(g) = \int_a^b g(t)dt$ is a linear functional.

2 Dual Space

We have seen previously that, for any two vector spaces V and W over F, $\mathscr{L}(V,W)$ is a vector space with respect to the pointwise operations

$$(T_1+T_2)(v) := T_1(v)+T_2(v)$$
$$(\alpha T)(v) := \alpha T(v).$$

Definition 2.1 (Dual space)**.** Let V be a vector space over F. We define the *dual space* of V, denoted by V^*, to be the vector space $V^* = \mathscr{L}(V,F)$.

Exercise 2.2. Show that if V is finite-dimensional, then $\dim V^* = \dim V$.

Recall the

Lemma 2.3 (Linear Map Lemma)**.** *Let V be a finite-dimensional vector space and W a vector space. Suppose $\{v_1,\dots,v_n\}$ is a basis of V and $w_1,\dots,w_n$ any n vectors in W. Then there exists a unique linear map $T : V \to W$ such that*

$$Tv_k = w_k$$

for each $k = 1,\dots,n$.

Suppose V is finite-dimensional and let $B = \{v_1,\dots,v_n\}$ be a basis of V. By the Linear Map Lemma, there exists a unique linear functional f_1 such that $f_1(v_1) = 1$ and $f_1(v_j) = 0$ for all $j = 2,\dots,n$. Similarly, there is a unique linear functional f_2 such that $f_2(v_2) = 1$ and $f_2(v_j) = 0$ for $j \neq 2$. Thus, for each $i = 1,\dots,n$, there exists a unique linear map f_i such that

$$f_i(v_j) = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}.$$

We can write this more simply as

$$f_i(v_j) = \delta_{ij},$$

where

$$\delta_{ij} := \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}$$

is called the *Kronecker delta*.

Exercise 2.4. For any $v \in V$, we can expand v in the ordered basis $B = (v_1, \ldots, v_n)$ as $v = \sum_{j=1}^{n} \alpha_j v_j$. Show that $f_i(v) = \alpha_i$, where f_i is the linear functional defined above. That is, the linear functional f_i picks out the coefficient of the ith basis vector v_i in the expansion of v with respect to B. Thus, we may also call f_i the *ith coordinate function for B*.

Exercise 2.5. Prove that the set of n linear functionals defined above, $B^* = \{f_1, \ldots, f_n\}$, is a basis of V^*. This basis is called the *dual basis* of the basis B of V.

Remark 2.6. In Exercise 1.2 we showed that every linear functional $f : F^n \to F$ is of the form $f(x_1, \ldots, x_n) = a_1 x_1 + \cdots + a_n x_n$, where $a_i = f(e_i)$. The result of the previous exercise generalizes this result. If V is a finite-dimensional vector space with $\dim V = n$, choose an ordered basis $B = (v_1, \ldots, v_n)$ and let $B^* = (f_1, \ldots, f_n)$ be the dual basis of V^*. Then any linear functional $f \in V^*$ can be written uniquely as

$$f = \sum_{i=1}^{n} f(v_i) f_i.$$

If $[v] = (x_1, \ldots, x_n)$ is the coordinate vector of v relative to B and we set $a_i = f(v_i)$, then

$$f(v) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i f_i \left(\sum_{j=1}^{n} x_j v_j \right) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i x_j f_i(v_j) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i x_j \delta_{ij} = \sum_{i=1}^{n} a_i x_i.$$

Exercise 2.7. Let $B = \{v_1, v_2, v_3\}$ be the basis of $\mathbb{R}^3$ defined by

$$v_1 = (1,0,-1), \quad v_2 = (1,1,1), \quad v_3 = (2,2,0).$$

Find the dual basis of $(\mathbb{R}^3)^*$.

3 Trace

An important linear functional on the space of $n \times n$ matrices is the *trace*.

Definition 3.1 (Trace of a matrix)**.** Define the *trace* of an $n \times n$ matrix A, denoted by $\text{tr}(A)$, to be the sum of the diagonal elements of A:

$$\text{tr}(A) = \sum_{i=1}^{n} A_{ii}.$$

Example 3.2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Then $\text{tr}(A) = 1+5+9 = 15$.

Exercise 3.3. Calculate $\text{tr}(I)$, where I is the $n \times n$ identity matrix.

Exercise 3.4. Show that $\text{tr} : M^{n\times n}(F) \to F$ is a linear functional.

Proposition 3.5 (Properties of the trace)**.** *Let A, B be $n \times n$ matrices and c a scalar. Then*

(i) $tr(A+B) = tr(A) + tr(B)$,

(ii) $tr(cA) = ctr(A)$,

(iii) $tr(A^T) = tr(A)$,

(iv) $tr(AB) = tr(BA)$,

Exercise 3.6. Prove Theorem 3.5.

Remark 3.7. It is straightforward to generalize property (iv) of Proposition 3.5 to any finite product of $n \times n$ matrices. For example, for three $n \times n$ matrices, property (iv) implies

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB).$$

This is called the *cyclic property* of the trace

Exercise 3.8. Prove that the trace of an $n \times n$ matrix A is a similarity invariant. That is, if A and B are similar $n \times n$ matrices, then they have the same trace.

Exercise 3.9. Prove that there do not exist $A, B \in M^{n \times n}(F)$ such that $AB - BA = I$.